



CSE 6220 High Performance Computing

Spring 2022

Midterm Exam I

February 21, 2022

NAME:

GT login:

Note:

- The exam has 5 questions for a total of 25 points. Please make sure that you have all the questions.
- Clearly state any assumptions you make in the solution of a problem.
- If the space provided is not sufficient, you may use additional sheets as necessary.

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Question	Max-Points	Points
1	5	
2	5	
3	5	
4	5	
5	5	
Total	25	

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1. (5 points) Answer the following questions (no justification required):

- (a) **True or False?** If $T(n, p)$ is the run-time of a parallel algorithm on p processors and $T(n, 1)$ is the run-time of the best sequential algorithm, then $T(n, 1) = O(pT(n, p))$.
- (b) **True or False?** Let $T(n, p_1)$ be the run-time of an algorithm on p_1 processors. For any $p_2 > p_1$, we can use the same algorithm to run in $\lceil \frac{p_1}{p_2} T(n, p_1) \rceil$ time on p_2 processors.
- (c) A parallel program yields a speedup of 24 when run on 32 processors. What is its efficiency at 32 processors?
- (d) **Yes or No?** Consider the operator $\otimes$ defined as $a \otimes b = 2a + b$. Can the operator be used with parallel prefix?
- (e) **Yes or No?** Consider the operator $\otimes$ defined as

$$a \otimes b = \begin{cases} b, & b \neq 0 \\ a, & \text{otherwise} \end{cases}$$

Can the operator be used with parallel prefix?

2. (5 points) The following algorithms are available for solving a problem of size n . A serial algorithm S that takes $\Theta(n^2)$ time, a parallel algorithm A_1 that takes $\Theta(n)$ time on n processors, and another parallel algorithm A_2 that takes $\Theta(\log n)$ time on n^2 processors.
- (a) Which parallel algorithm is more efficient? (2 points)
 - (b) Which parallel algorithm gives you maximum speed? (1 point)
 - (c) Which algorithm should be recommended for practical use? Is there a clear choice? (1 point)
 - (d) Now, suppose A_1 consumes $10n$ bytes of memory and A_2 consumes n^2 bytes of memory. Taking this new information into account, which algorithm should be recommended? (1 point)

3. (5 points) The best sequential and parallel algorithms for solving a problem have the following run-times:

$$T(n, 1) = \Theta(n)$$

$$T(n, p) = \Theta\left(\frac{n}{p} + \sqrt{p} \log p\right)$$

What is the maximum number of processors (as a function of n) that can be used while still achieving optimal efficiency?

4. (5 points) A sequence of nested parenthesis is said to be well-formed if 1) there are an equal number of left and right parenthesis and 2) each right parenthesis is matched by a left parenthesis that occurs to its left in the sequence. For example, $((()())())$ is well-formed but $()()()$ is not.

There is a nested parenthesis sequence of length n distributed across p processors using block decomposition. Design a parallel algorithm to determine if it is well-formed and find its run-time.

5. (5 points) Consider an array A of n numbers ($A[0] \dots A[n-1]$). For each entry of A that is 0, we want to replace it with the nearest non-zero entry to the left, i.e., for each $A[i] = 0$ we want to replace it with $\max_{j < i} A[j] \neq 0$. Design a parallel algorithm to solve this problem and compute its run-time. You may assume the first entry of A is non-zero.